

BAYREM BEN RAYEN

AIops Specialist | AI & Machine Learning Engineer

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PROFESSIONAL SUMMARY

Recent Software Engineering graduate specializing in AI-driven infrastructure and AIops. Built and shipped an AI-powered server monitoring platform combining anomaly detection, predictive forecasting, and LLM-generated incident reporting. Strong hands-on foundation in Python, machine learning (Scikit-Learn, TensorFlow), and self-hosted infrastructure (Docker, PostgreSQL, workflow automation), with a focus on turning operational data into automated, actionable insight.

SKILLS & TECHNOLOGIES

Languages: Python (Expert), Java (Advanced), Angular (Intermediate)
AI / ML: Scikit-Learn, TensorFlow, Isolation Forest, K-Means Clustering, Linear Regression, Ollama / LLaMA 3, Prompt Engineering
Data Tools: Pandas, Matplotlib, Joblib, Data Mining & Preparation
Infrastructure / AIops: Docker, FastAPI, PostgreSQL, Grafana, n8n (workflow automation), Linux, Flask
Soft Skills: Critical Thinking (Advanced), Problem Solving (Advanced), Time Management (Advanced)

PROFESSIONAL EXPERIENCE

- End-of-Studies Internship — AIops / AI Infrastructure — TriTUX Group***Feb 2026 – Jun 2026*
 - Designed and built an AI-powered server monitoring platform (Net Detective) delivering real-time anomaly detection, predictive forecasting, and automated incident reporting.
 - Shipped Isolation Forest anomaly detection and linear regression forecasting as production FastAPI endpoints, exposed to n8n automation workflows.
 - Automated daily plain-English incident reports via an LLM (Ollama / LLaMA 3), delivered to stakeholders through Telegram.
 - Built n8n workflows for 1-minute threshold alerting and 5-minute anomaly/predictive checks; resolved alert-spam issues with a cooldown mechanism.
- Data Mining & Machine Learning Intern — Tunisie Telecom***Jul 2025 – Sep 2025*
 - Designed and developed a full customer prediction solution delivered as a Flask web application.
 - Performed data mining and preparation with advanced Pandas manipulation and Matplotlib visualization.
 - Implemented K-Means clustering for unsupervised customer segmentation.
 - Deployed the optimized prediction model for web integration using Joblib.

KEY PROJECTS

- Net Detective — AI-Powered Server Monitoring Platform**
Stack: Docker, FastAPI, PostgreSQL, Grafana, n8n, Ollama/LLaMA 3, Scikit-Learn
 - End-to-end AIops platform combining anomaly detection, predictive forecasting, and automated LLM-based reporting for server infrastructure, developed across 3 Scrum sprints in a 3-person team.
 - Built /detect and /predict FastAPI endpoints for real-time anomaly detection and forecasting; produced full architecture and UML sequence diagrams.
 - Diagnosed and resolved production issues including SQL operator-precedence bugs and n8n HTTP request serialization.

EDUCATION

- Bachelor's Degree (Licence) — Software Engineering & Information Systems — École Centrale Supérieure Privée d'Informatique et de Télécommunications, Honoris United Universities***Sept 2023 – Jul 2026*
- Baccalauréat — Computer Science — Lycée Mohamed Attia Khaznadar***Sept 2022 – Sept 2023*

CERTIFICATIONS

- HCIA-AI V4.0 Course** , Huawei ICT Academy, Aug 2026 [\[certificate\]](#)
- Python Essentials 2** — Cisco Networking Academy, Dec 2025
- Hashgraph Developer** — DarBlockchain, Jun 2025 [\[credential\]](#)
- Prompt Engineering — 21st Century Skills** — Honoris United Universities, May 2025
- Entrepreneurship — 21st Century Skills** — Honoris United Universities, May 2025
- Personal Development — 21st Century Skills** — Honoris United Universities, May 2025
- Coding and Data Analytics** — Le Wagon, May 2023

LANGUAGES

Languages: Arabic (Native), English (C2), French (B2), German (B1)

INTERESTS

Road-trips, Music, Gaming, Innovation, AI Training